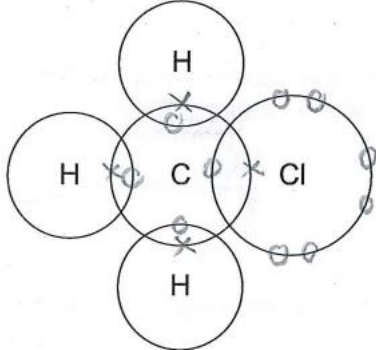


| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                   | Marks available |     |     |       |       |      |
|----------|-----|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                   | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 4        | (a) |  |  | transfer of one electron from calcium to each chlorine atom (1)<br>can be implied by electrons being in correct place in chloride ions<br><br>octet in each chloride ion (1)<br>ignore octet included in calcium ion<br><br>correct charge on all ions – Ca <sup>2+</sup> and Cl <sup>-</sup> (1) |                 | 3   |     | 3     |       |      |
|          | (b) |  |  | <br><br>four shared pairs (1)<br><br>octet around Cl atom (1)                                                                                                                                                    |                 | 2   |     | 2     |       |      |